PREDICTING CATALOG DEMAND REPORT

Part 2 - Lesson 5 Project

ABSTRACT

Throughout this report I will demonstrate every taken step to help a mail-order catalog business company to make a right decision regarding send this year's catalog out to their 250 new customers or do not.

Maged Baheig

Problem Solving with Advanced Analytics-Udacity/Bertelsmann2021

Predicting Catalog Demand Report

<u>Step 1</u>: Business and Data Understanding

Provide an explanation of the key decisions that need to be made. (500 word limit)

Key Decisions:

Answer these questions

1. What decisions needs to be made?
2. What data is needed to inform those decisions?

A **mail-order catalog business company** –that manufactures and sells high-end home goods– is preparing to send out **this year's catalog** in the **coming months**. The company need a help to make a right decision regarding **send** this year's catalog out to their 250 new customers or **not**. The sending decision will be made based on the **expected profit contribution**, it must **exceed $10,000** to send the catalog.

- I will use a large database ***p1-customers.xlsx*** *(about 2375 Records*) contains some customer details and transactional info to build a model to predict the **expected profit contribution** based on these customer and transactional info.
- <u>Based on my model result</u>, I will use my model and ***p1-mailinglist.xlsx*** *(250 Records*) estimate the **expected profit contribution** the company may earn if send the catalog to their 250 customers.
- Management will **send** the catalog **only** if the **expected profit** contribution **exceeds $10,000**.

	A	B	C	D	E	F	G	H	I	J	K	L
1	Name	Customer_Segment	Customer_ID	Address	City	State	ZIP	Avg_Sale_Amount	Store_Number	Responded_to_Last_Catalog	Avg_Num_Products_Purchased	#_Years_as_Customer
2	Pamela Wright	Store Mailing List	2	376 S Jasmine St	Denver	CO	80224	227.9	100	No	1	6
3	Danell Valdez	Store Mailing List	7	12066 E Lake Cir	Greenwood Village	CO	80111	55	105	Yes	1	6
4	Jessica Rinehart	Store Mailing List	8	7225 S Gaylord St	Centennial	CO	80122	212.57	101	No	1	3
5	Nancy Clark	Store Mailing List	9	4497 Cornish Way	Denver	CO	80239	195.31	105	Yes	1	6
6	Andrea Brun	Store Mailing List	10	2316 E 5th Ave	Denver	CO	80206	110.55	100	Yes	1	2
7	Denise Pentico	Store Mailing List	11	3883 Quitman St	Denver	CO	80212	149.01	106	No	1	8
8	Erna Arustamyan	Store Mailing List	12	1965 Yukon St	Lakewood	CO	80214	49.37	108	No	1	7
9	Karen Osborne	Store Mailing List	16	5400 Sheridan Blvd	Arvada	CO	80002	153.97	103	No	3	1
10	Shirley Krywonis	Store Mailing List	17	195 Jade St	Broomfield	CO	80020	173.15	107	No	2	2
11	Dianne Vangilder	Store Mailing List	19	22873 E Clifton Pl	Aurora	CO	80016	105.24	102	No	1	6
12	Mary Kiniry	Store Mailing List	26	16087 E Lehigh Cir	Aurora	CO	80013	245.16	104	No	4	3
13	Belinda Deherrera	Store Mailing List	28	14303 E Napa Pl	Aurora	CO	80014	85.02	102	No	2	1

A Screenshot of p1-customers.xlsx

	A	B	C	D	E	F	G	H	I	J	K	L
1	Name	Customer_Segment	Customer_ID	Address	City	State	ZIP	Store_Number	Avg_Num_Products_Purchased	#_Years_as_Customer	Score_No	Score_Yes
2	A Giametti	Loyalty Club Only	2213	5326 S Lisbon Way	Centennial	CO	80015	105	3	0.2	0.694964193	0.305035807
3	Abby Pierson	Loyalty Club and Credit Card	2785	4344 W Roanoke Pl	Denver	CO	80236	101	6	0.6	0.527275463	0.472724537
4	Adele Hallman	Loyalty Club Only	2931	5219 S Delaware St	Englewood	CO	80110	101	7	0.9	0.42111815	0.57888185
5	Alejandra Baird	Loyalty Club Only	2231	2301 Lawrence St	Denver	CO	80205	103	2	0.6	0.694862189	0.305137811
6	Alice Dewitt	Loyalty Club Only	2530	5549 S Hannibal Way	Centennial	CO	80015	104	4	0.5	0.612294145	0.387705855
7	Amanda Donahoe	Credit Card Only	1946	10093 E Warren Ave	Denver	CO	80247	105	7	0.7	0.732721707	0.267278293
8	Amanda Huerta	Loyalty Club and Credit Card	1212	3889 Aldenbridge Cir	Highlands Ranch	CO	80126	101	4	1	0.778260506	0.221739494

A Screenshot of p1-mailinglist.xlsx

Key Decisions:

1. **Decision**: Whether the company **should Send the catalog to the 250 customers or should not**, based on "***expected profit criteria***".

2. **Needed Info**: The **expected profit if catalog will be sent**.

3. **Analysis Type**: **Predicative** analysis to **predict the expected profit** from sending catalog action.

4. **Needed Data:** any variables (**critical factors**) may affect customer **purchase decision** and thus affect the "**Expected Profit**".

<u>Step 2</u>: Analysis, Modeling, and Validation

Provide a description of how you set up your linear regression model, what variables you used and why, and the results of the model. Visualizations are encouraged. (500 word limit)

Important: Use the p1-customers.xlsx to train your linear model.

At the minimum, answer these questions:

1. How and why did you select the predictor variables in your model? You must explain how your continuous predictor variables you've chosen have a linear relationship with the target variable. Please refer back to the "Multiple Linear Regression with Excel" lesson to help you explore your data and use scatterplots to search for linear relationships. You must include scatterplots in your answer.

2. Explain why you believe your linear model is a good model. You must justify your reasoning using the statistical results that your regression model created. For each variable you selected, please justify how each variable is a good fit for your model by using the p-values and R-squared values that your model produced.

3. What is the best linear regression equation based on the available data? Each coefficient should have no more than 2 digits after the decimal (ex: 1.28)

Important: The regression equation should be in the form:

*Y = Intercept + b1 * Variable_1 + b2 * Variable_2 + b3 * Variable_3……*

For example: Y = 482.24 + 28.83 * Loan_Status – 159 * Income + 49 (If Type: Credit Card) – 90 (If Type: Mortgage) + 0 (If Type: Cash)

Note that we **must** include the 0 coefficient for the type Cash.

Note: For students using software other than Alteryx, if you decide to use Customer Segment as one of your predictor variables, please set the base case to Credit Card Only.

■ Analysis

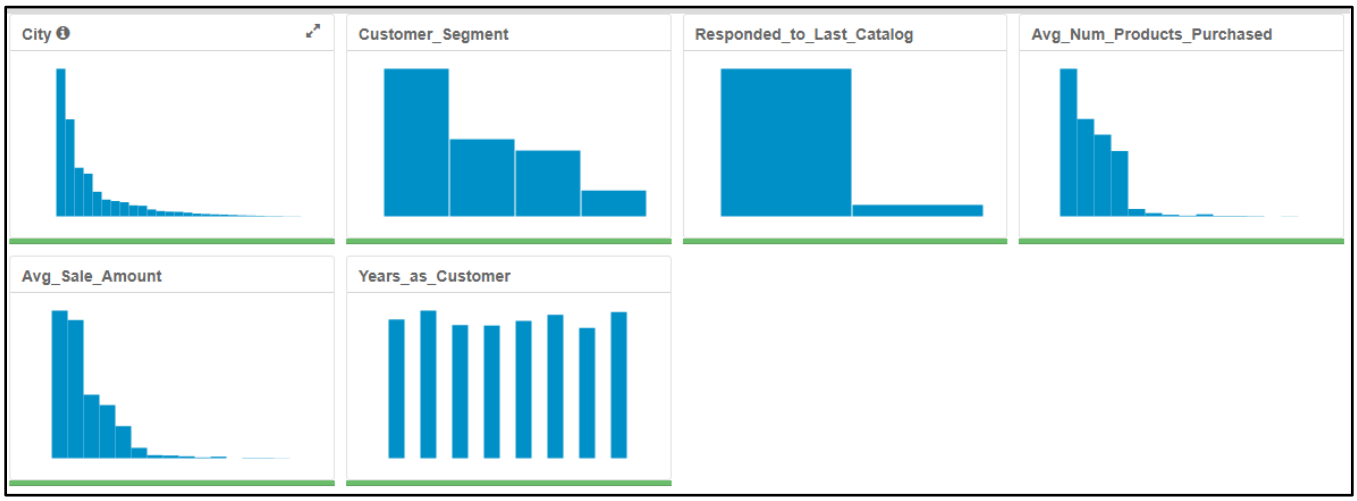
- <u>I used the dataset at **p1-customers.xlsx** file to</u>:
  1. Exploring its values to build **assumption** about **Critical Factors** may affect customer **purchase decision** and thus affect the "**Expected Profit**". <u>I suspected below **4 variables**</u>:
     - Customer_Segment.
     - City.
     - Avg_Num_Products_Purchased
     - Years_as_Customer.

2. Testing my assumption about the **potential Critical Factors** using **Scatter Plots** and **SLR**.

- <u>Done Steps</u>:

1. Exploring p1-customers.xlsx values

- **No Missing Values**. No need for extra data or cleansing.

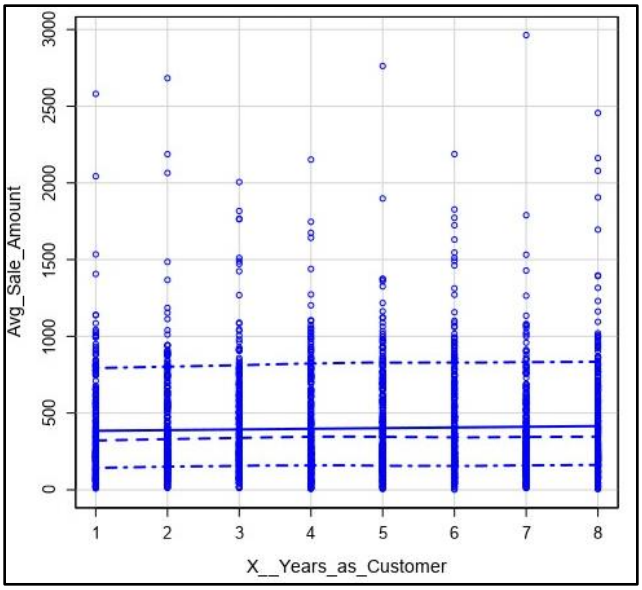


2. Exploring p1-mailinglist.xlsx values

- **No Missing Values**. No need for extra data or cleansing.

3. Checking Years_as_Customer/Avg_Sale_Amount Relation

a. Scatter Plot



b. Simple Linear Regression Result

Report

Report for Linear Model Years_Sales_SLR

Basic Summary

Call:
lm(formula = Avg_Sale_Amount ~ X._Years_as_Customer, data = the.data)

Residuals:

Min	1Q	Median	3Q	Max
-410	-233	-119	174	2553

Coefficients:

	Estimate	Std. Error	t value	Pr(>|t|)
(Intercept)	380.039	15.283	24.867	< 2.2e-16 ***
X._Years_as_Customer	4.385	3.021	1.451	0.14679

Significance codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 340.04 on 2373 degrees of freedom
Multiple R-squared: 0.000887, Adjusted R-Squared: 0.0004659
F-statistic: 2.107 on 1 and 2373 degrees of freedom (DF), p-value 0.1468

Type II ANOVA Analysis

Response: Avg_Sale_Amount

	Sum Sq	DF	F value	Pr(>F)
X._Years_as_Customer	243578.02	1	2.11	0.14679
Residuals	274377805.08	2373		

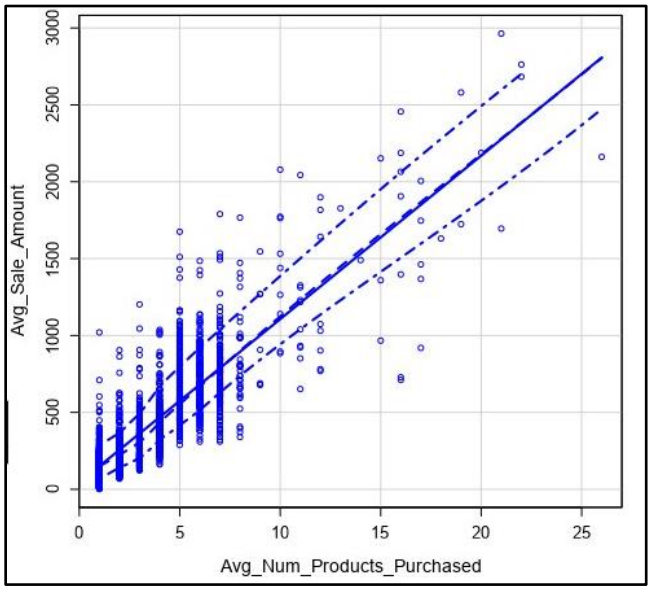
Significance codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

c. <u>Conclusion</u>:

- Based on scatter plot shape, **high** P-value (**14.7%**), and **low** R-squared (**0.0009**), Years_as_Customer is **NOT** a **good predictor** for Avg_Sals_Amount. Neither if used in **SLR** nor **MLR**.

4. Checking Avg_Num_Products_Purchased/Avg_Sale_Amount Relation

a. *Scatter Plot*



b. *Simple Linear Regression Result*

Report

Report for Linear Model Product_Sales_SLR

Basic Summary

Call:
lm(formula = Avg_Sale_Amount ~ Avg_Num_Products_Purchased, data = the.data)

Residuals:

Min	1Q	Median	3Q	Max
-1033.2	-99.4	-17.8	71.4	1099.4

Coefficients:

	Estimate	Std. Error	t value	Pr(>|t|)
(Intercept)	44.02	5.704	7.716	1.75e-14 ***
Avg_Num_Products_Purchased	106.28	1.319	80.572	< 2.2e-16 ***

Significance codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 176.01 on 2373 degrees of freedom
Multiple R-squared: 0.7323, Adjusted R-Squared: 0.7322
F-statistic: 6492 on 1 and 2373 degrees of freedom (DF), p-value < 2.2e-16

Type II ANOVA Analysis

Response: Avg_Sale_Amount

	Sum Sq	DF	F value	Pr(>F)
Avg_Num_Products_Purchased	201109435.07	1	6491.91	< 2.2e-16 ***
Residuals	73511948.03	2373		

Significance codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

c. <u>Conclusion</u>:

- Based on scatter plot shape (**Positive Relationship**), **low** P-value (**2.2E-16**), and **high** R-squared (**0.73**), Avg_Num_Products_Purchased is a **good predictor** for Avg_Sals_Amount.
- It's also important to mention that, it ***can explain or predict only 73% of our dataset variation***. So, *it would be great if I found another predictor variable(s) can explain more beside it*.

■ Model and Validation

- <u>Based on initial analysis, I decided to do the following</u>:

  1. Build a **Multiple Linear Regression model** to predict the expected profit if catalog sent, based on:
     - **Customer_Segment**.
     - **City**.
     - **Avg_Num_Products_Purchased**
     - **Years_as_Customer**.

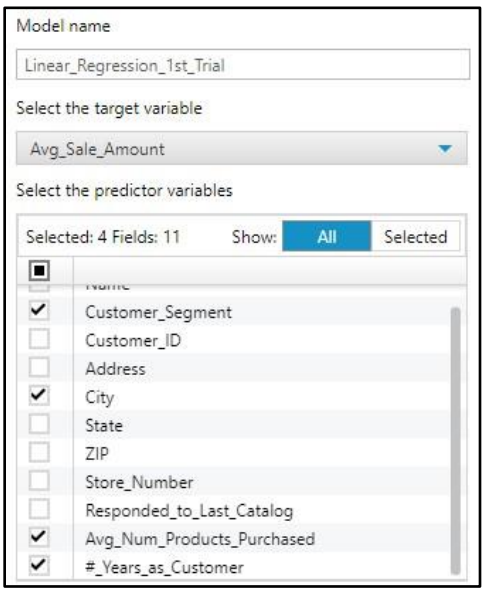
  2. Validate that model:
     - <u>If the 4 selected predictor variables are **significant** with **good R-Squared**</u>, I will use that model.
     - <u>**If not**</u>, I will exclude the **insignificant** variables(s) and **rebuild** the model.

  3. Setup the **best linear regression equation** based on the best model and use this equation to predict the expected profit if catalog sent to the new 250 Customers.

- <u>Created Models</u>:

  1. 1st Model:

  a. Used dataset



b. Multiple Linear Regression Result

Coefficients:

	Estimate	Std. Error	t value	Pr(>|t|)
(Intercept)	317.70206	14.531	21.864191	< 2.2e-16 ***
Customer_SegmentLoyalty Club Only	-149.40802	9.016	-16.571610	< 2.2e-16 ***
Customer_SegmentLoyalty Club and Credit Card	282.75358	11.973	23.616745	< 2.2e-16 ***
Customer_SegmentStore Mailing List	-245.02944	9.843	-24.894717	< 2.2e-16 ***
CityAurora	-15.95571	10.734	-1.486437	0.1373
CityBoulder	-45.19755	80.071	-0.564472	0.57249
CityBrighton	-70.26566	97.691	-0.719264	0.47205
CityBroomfield	-5.30736	15.109	-0.351263	0.72542
CityCastle Pines	-84.44275	97.670	-0.864575	0.38736
CityCentennial	-5.25704	17.886	-0.293917	0.76885
CityCommerce City	-33.07574	44.476	-0.743671	0.45715
CityDenver	3.98511	10.095	0.394771	0.69305
CityEdgewater	29.83702	40.665	0.733719	0.46319
CityEnglewood	10.77182	20.369	0.528845	0.59696
CityGolden	-12.77509	32.762	-0.389937	0.69662
CityGreenwood Village	-45.22426	37.899	-1.193270	0.23288
CityHenderson	-286.00090	138.045	-2.071802	0.03839 *
CityHighlands Ranch	-20.34032	30.014	-0.677685	0.49804
CityLafayette	-44.42012	62.177	-0.714412	0.47504
CityLakewood	-6.22696	12.815	-0.485911	0.62708
CityLittleton	-22.24567	18.423	-1.207475	0.22737
CityLone Tree	81.03573	137.947	0.587440	0.55696
CityLouisville	-30.95156	69.344	-0.446346	0.65539
CityMorrison	-12.33847	52.749	-0.233911	0.81507
CityNorthglenn	-16.05329	29.430	-0.545480	0.58548
CityParker	0.03713	27.891	0.001331	0.99894
CitySuperior	-55.85242	46.710	-1.195738	0.23192
CityThornton	28.83163	24.848	1.160329	0.24603
CityWestminster	-7.01672	17.309	-0.405386	0.68523
CityWheat Ridge	6.25759	20.682	0.302566	0.76225
Avg_Num_Products_Purchased	67.18223	1.527	44.002385	< 2.2e-16 ***
X._Years_as_Customer	-2.35723	1.233	-1.912298	0.05596 .

Significance codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 137.53 on 2343 degrees of freedom
Multiple R-squared: 0.8386, Adjusted R-Squared: 0.8365
F-statistic: 392.8 on 31 and 2343 degrees of freedom (DF), p-value < 2.2e-16

Type II ANOVA Analysis

Response: Avg_Sale_Amount

	Sum Sq	DF	F value	Pr(>F)
Customer_Segment	28366598.36	3	499.89	< 2.2e-16 ***
City	409700.73	26	0.83	0.70657
Avg_Num_Products_Purchased	36623567.35	1	1936.21	< 2.2e-16 ***
X._Years_as_Customer	69170.27	1	3.66	0.05596 .
Residuals	44318035.67	2343		

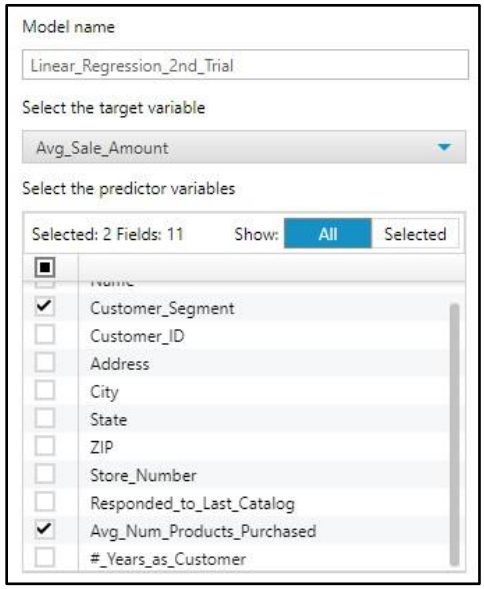
Significance codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

c. <u>**Conclusion**</u>:

- **High Adjusted R-Squared** (**0.84**).
- **Low P-Values** for the **2** attributes (*Customer_Segment & Avg_Num_Products_Purchased*).
- **High P-Values** for the **2** attributes (*City & Years_as_Customer*).
- **<u>I will rebuild the model</u>**, after excluding (*Customer_Segment & Avg_Num_Products_Purchased*).

2. 2nd Model:

a. *Used dataset*



b. *Multiple Linear Regression Result*

Report for Linear Model Linear_Regression_2nd_Trial

Basic Summary

Call:
lm(formula = Avg_Sale_Amount ~ Customer_Segment + Avg_Num_Products_Purchased, data = the.data)

Residuals:

Min	1Q	Median	3Q	Max
-663.8	-67.3	-1.9	70.7	971.7

Coefficients:

	Estimate	Std. Error	t value	Pr(>|t|)
(Intercept)	303.46	10.576	28.69	< 2.2e-16 ***
Customer_SegmentLoyalty Club Only	-149.36	8.973	-16.65	< 2.2e-16 ***
Customer_SegmentLoyalty Club and Credit Card	281.84	11.910	23.66	< 2.2e-16 ***
Customer_SegmentStore Mailing List	-245.42	9.768	-25.13	< 2.2e-16 ***
Avg_Num_Products_Purchased	66.98	1.515	44.21	< 2.2e-16 ***

Significance codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 137.48 on 2370 degrees of freedom
Multiple R-squared: 0.8369, Adjusted R-Squared: 0.8366
F-statistic: 3040 on 4 and 2370 degrees of freedom (DF), p-value < 2.2e-16

Type II ANOVA Analysis

Response: Avg_Sale_Amount

	Sum Sq	DF	F value	Pr(>F)
Customer_Segment	28715078.96	3	506.4	< 2.2e-16 ***
Avg_Num_Products_Purchased	36939582.5	1	1954.31	< 2.2e-16 ***
Residuals	44796869.07	2370		

Significance codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

c. <u>Conclusion</u>:

- **High Adjusted R-Squared** (**0.84**).
- **Low P-Values** for the **2** attributes (*Customer_Segment & Avg_Num_Products_Purchased*).

- Based on above analysis, model and validation, the **best linear regression equation** is:

Y (Predicted Sales Amount) = 303.46 +

(-149.36 * Customer_SegmentLoyalty Club Only) +
(281.84 * Customer_SegmentLoyalty Club and Credit Card) +
(-245.42 * Customer_SegmentStore Mailing List)
+ 0 (If Type: Customer_SegmentCredit Card Only) +
(66.98 * Avg_Num_Products_Purchased)

Step 3: Presentation/Visualization

Use your model results to provide a recommendation. (500 word limit)
At the minimum, answer these questions:

1. What is your recommendation? Should the company send the catalog to these 250 customers?

2. How did you come up with your recommendation? (Please explain your process so reviewers can give you feedback on your process)

3. What is the expected profit from the new catalog (assuming the catalog is sent to these 250 customers)?

Recommendation

- I encourage this company to **send this year's catalog** to the new **250** customers, as it is expected to gain (**21987.44 $**) as Net Profit, which is more than the set criteria (**10000 $**).

Process Summary and Justification

- As mentioned above, I **recommend** a bid of **21987.44 $.** That mainly comes from my analysis which ended up with below **best model result**:

Residual standard error: 137.48 on 2370 degrees of freedom
Multiple R-squared: 0.8369, Adjusted R-Squared: 0.8366
F-statistic: 3040 on 4 and 2370 degrees of freedom (DF), p-value < 2.2e-16

Type II ANOVA Analysis

Response: Avg_Sale_Amount

	Sum Sq	DF	F value	Pr(>F)
Customer_Segment	28715078.96	3	506.4	< 2.2e-16 ***
Avg_Num_Products_Purchased	36939582.5	1	1954.31	< 2.2e-16 ***
Residuals	44796869.07	2370		

Significance codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

- I **arrived** at this number by using the **linear equation formula** from the **2nd regression model** I made based on previous customer attributes and average sales amount from those customers who responded to last year catalog.

- Then, and I **applied** it to the dataset which contains **250 new customers** whom the company intent to send them the new catalog *if the expected profit from sending exceeds **10,000$**.* I got **"Predicted Sales Amount"** for each customer as a result from applying the formula.

- Then, I **multiplied** the **probability** that a person will buy the catalog as well to each **"predicted sales amount result"** for each customer. **Summed** the result to get **"Expected Revenue"**.

- Then, I **factored** in average **gross margin 50%** so I **multiply** the "**Expected Revenue**" **47224.87$** by **.50**, and **subtract** out the printing and distributing per catalog **($6.50 * 250 customers)** to get the final **"Expected Profit"** of **21987.44$**.

Thank You